PRANITA TASHILDAR

Troy, MI | (248) 315-1779 | ptashild@mail.walshcollege.edu | linkedin.com/in/pranita-tashildar | github.com/Pranita2798

Data Analyst & AI/ML Engineer with 5+ years transforming data into business results through statistical analysis, predictive modeling, and AI-powered automation. Expert in Python, SQL, Power BI, and machine learning with hands-on experience in RAG architectures, LLM integration (OpenAI, Claude), and cloud deployment (GCP, AWS). Track record: 85% error reduction, 22% forecasting improvement, $200K+ cost savings, 45% efficiency gains, 60% automation improvement. MS in Data Analytics with cross-functional experience across Fortune 500 consulting, automotive technology, and enterprise applications.

TECHNICAL SKILLS

Data Analysis: Python (Pandas, NumPy, Scikit-learn), SQL (MySQL, PostgreSQL), R, Statistical Analysis, Predictive Modeling, Forecasting, Machine Learning

AI/ML: LangChain, LlamaIndex, Hugging Face, RAG Architecture, OpenAI API, Claude API, Vector Databases (FAISS, Pinecone), Prompt Engineering, NLP

Business Intelligence: Power BI (DAX, Star Schema), Tableau, Grafana, Dashboard Development, Data Visualization, Executive Reporting

Cloud & Tools: GCP, AWS, ETL Pipelines, Data Governance, n8n Automation, Git, Excel (VBA, Power Query), REST APIs

PROFESSIONAL EXPERIENCE

Software Associate & AI/ML Engineer | Methodica Technologies | Troy, MI | 2025

- Developed AI-powered IT ticketing automation using OpenAI API to classify incidents by priority, auto-routing 78% of tickets and reducing SLA resolution time 45%
- Architected RAG system using LangChain and Hugging Face with FAISS/Pinecone vector stores for technical documentation Q&A, achieving 95%+ retrieval accuracy
- Implemented ML anomaly detection for automotive sensor data using unsupervised learning, eliminating 85% data corruption and generating $200K+ annual savings
- Designed predictive monitoring dashboards using Grafana and InfluxDB, accelerating incident response time by 35% through real-time alerting
- Created automated test case generator using OpenAI API with RAG architecture, streamlining manual creation by 60% and conserving 120+ hours monthly

Graduate Teaching Assistant | Walsh College | Troy, MI | 2024-2025

- Instructed 25+ students in Python, R, Power BI, Tableau, and machine learning, improving project outcomes by 18% with 92% satisfaction
- Developed curriculum covering statistical modeling, predictive analytics, ETL design, and dashboard development

Data Analyst | Bright Consultancy Services | Pune, India | 2024

- Constructed machine learning models for revenue forecasting using Python and SQL, improving prediction accuracy by 22% and enabling $2M+ strategic planning for Fortune 500 clients
- Established Power BI dashboards with DAX and star schema design, reducing planning cycles by 3 weeks through automated reporting
- Streamlined reporting workflows using Python scripts and ETL processes, decreasing manual effort by 45% across 8 concurrent client projects

Data Analyst | Shree Samarth Constructions | Belgaum, India | 2020-2023

- Reduced material waste by 22% (saving $150K+ annually) through predictive modeling analyzing usage patterns across $15M+ projects
- Established MySQL databases and Power BI dashboards for real-time resource allocation, expediting project delivery by 15%
- Coordinated with contractors, suppliers, and regulatory agencies, automating reporting and eliminating 50% manual tracking

KEY PROJECTS

Enterprise RAG & AI Automation System

- Engineered LangChain/LlamaIndex with FAISS/Pinecone for semantic search, achieving 95%+ retrieval accuracy with multi-model routing (GPT-4, Claude)
- Deployed conversational AI chatbots on AWS integrating OpenAI and Claude APIs with n8n workflows, processing 500+ daily queries with 89% resolution rate

Healthcare Analytics & Predictive Modeling

- Conducted statistical analysis achieving 20% cost reduction while maintaining 95% efficacy using Python and machine learning
- Forecasted patient demand and optimized resource allocation, creating visualization dashboards for stakeholder KPI monitoring

Automotive Sensor Data Pipeline (DataRack Pro)

- Constructed end-to-end ML system processing gigabytes of sensor data (radar, camera, GNSS, LiDAR) daily with 99.9% uptime on GCP
- Implemented ML anomaly detection and predictive SSD health monitoring, attaining 85% error reduction with automated S3 uploads

EDUCATION

Master of Science in Data Analytics | Walsh College, Troy, MI | 2025

Coursework: Machine Learning, Predictive Analytics, Statistical Modeling, Business Intelligence, Data Mining

Bachelor of Engineering in Electronics & Communication | India | 2023